

Ch. 6 Index of a Severi-Brauer Variety

ref: [GS] Gille, Szamuely

SB var

defn

- index of X
 $\text{ind}(X) := \gcd(\text{degrees of all } 0\text{-cycles on } X)$
- period of X
 $\text{per}(X) := \text{order of } [X] \text{ in } \text{Br}(k)$

CSA

defn

- index of A
 $\text{ind}(A) := \gcd([K:k] \mid K/k: \text{fin. sep. s.t. split } A)$
- period of A
 $\text{per}(A) := \text{order of } [A] \text{ in } \text{Br}(k)$

Rem "index of a variety" has various meanings depending on the case.

- Esnault - Levine - Wittenberg : today! c.f. Kollár "ELW-indices"
- Weil : the smallest natural number n s.t. nK_X Cartier
- Cartier / Fano : the largest natural number m s.t. $\frac{1}{m}K_X$ Cartier

Property

P : SB var

$P^{\min}$: minimal twisted linear subvar

$$\text{ind}(P) = \text{ind}(P^{\min}) = \dim P^{\min} + 1$$

Thm 53

Property

$\exists$ by Wedderburn

A : CSA

D : division alg s.t. $A \cong M_r(D)$

$$\text{ind}(A) = \text{ind}(D) \cdot \deg(D)$$

[GS] Prop 4.5.8.

chatlet ([GS] Cor 5.3.6)

$$\text{ind}(P) = \dim P^{\min} + 1 = \text{ind}(A)$$

• P contains sm 0-dim sub scheme of degree $\text{ind}(P)$ (Thm 53)

(i.e. P has K -pt for some separable field ext. K/k of degree $\text{ind}(P)$)

• D contains a subfield K which is of degree $\text{ind}(D)$ s.t. D splits over K ([GS] Prop 4.5.3)

• P, Q generate the same subgr of $\text{Br}(k)$ then $\text{ind}(P) = \text{ind}(Q)$ (Cor 52)

• A, B generate the same subgr⁽²⁾ of $\text{Br}(k)$, then $\text{ind}(A) = \text{ind}(B)$ ([GS] Cor 4.5.10)

• $\text{ind}(P) = 1 \iff P$ has k -rat pt
 $\iff P \cong P_k^{n-1}$

• $\text{ind}(A) = 1 \iff A$ split
 $\iff A \cong M_n(k)$

(Châtelet [GS] Thm 5.1.3)

([GS] Rem 4.5.2)

Note • in general

X has k -rat. pt $\Rightarrow \text{ind}(X) = 1$
 However, the converse does not hold in general.

e.g. $p > 3$, $f(x, y) = x^{p-1} + y^{p-1} + 1$, $X = (f=0)$

we have $X(\mathbb{F}_p) = \emptyset$

• Classifying of varieties X which has the property

$\text{ind}(X) = 1 \Rightarrow X(k) \neq \emptyset$

is open (cf: Colliot-Thélène "The index of alg. var")

ind and per

• $\text{per}(P) \mid \text{ind}(P)$

• $\text{per}(P)$ & $\text{ind}(P)$ have the same prime factors (Cor 54)

ind and per

• $\text{per}(A) \mid \text{ind}(A)$

• $\text{per}(A)$ & $\text{ind}(A)$ have the same prime factors (Brauer [GS] Prop 4.5.13)

index-period Problem

asks to determine the smallest number m (depending on k) such that $\text{ind}(P) \mid \text{per}(P)^m$

recent work $\rightarrow \dots$

Primary decomp. (Prop 57)

P : minimal SB var

$\text{ind}(P) = \prod_i p_i^{c_i}$ (p_i : distinct primes)

$\Rightarrow P \cong \bigotimes_i P_i$ (P_i : min SB var)

s.t. $\text{ind}(P_i) = p_i^{c_i}$

Primary decomp. (Brauer [GS] Prop 4.5.16)

D : central division alg

$\text{ind}(D) = \prod_i p_i^{c_i}$ (p_i : distinct primes)

$\Rightarrow D \cong \bigotimes_i D_i$ (D_i : cent. div. alg)

s.t. $\text{ind}(D_i) = p_i^{c_i}$

Proofs

Thm 53 Let P be a SB var, then.

$$(1) \text{ ind}(P) = \text{ind}(P^{\min})$$

$$(2) \text{ ind}(P) = \dim P^{\min} + 1$$

(3) P contains sm. 0-dim sub scheme of degree $\text{ind}(P)$.

Pf By Cor 30 (Session 5)

$$P \cong P^{\min} \times \mathbb{P}^c \quad (c = \dim P - \dim P^{\min})$$

hence any 0-cycle on P (resp. $P^{\min}$)

yields a 0-cycle of the same degree on $P^{\min}$ (resp. P)

Then $\text{ind}(P) \mid \text{ind}(P^{\min})$ and $\text{ind}(P^{\min}) \mid \text{ind}(P)$, then (1) holds.

By Lem 19 (Session 4),

(idea: a general section of the tangent bundle of $\mathbb{P}^m$ has $m+1$ zeros)

$P^{\min}$, and hence P , contain sm. 0-dim subscheme of degree $\dim P^{\min} + 1$, thus $\text{ind}(P) \leq \dim P^{\min} + 1$.

Claim

the degree of every 0-dim. reduced subscheme $Z \subset P$ is divisible by $\dim P^{\min} + 1$

Set $d := \dim P^{\min} + 1$ and choose $r > 0$ s.t. $rd + 1 \geq \deg Z$

then on $P_{\bar{k}} \cong \mathbb{P}_{\bar{k}}^{n-1}$ there exists a hyper surface of degree $(rd+1)$ passing through the support of $Z_{\bar{k}}$

$$\text{i.e. } H^0(P_{\bar{k}}, \mathcal{O}_{P_{\bar{k}}}(rd+1)) \rightarrow H^0(Z_{\bar{k}}, \mathcal{O}_{P_{\bar{k}}}(rd+1)|_{Z_{\bar{k}}}) \cong H^0(Z_{\bar{k}}, \mathcal{O}_{Z_{\bar{k}}}) \cong \bar{k}^{\deg Z}$$

Its kernel defines a twisted linear sub var.

$$|\mathcal{O}_P(rd+1)|(-Z) \subset |\mathcal{O}_P(rd+1)| \quad \text{of codim } \deg Z$$

Note • $\mathcal{O}_P(d)$ is a line bundle on P (Prop 44 (Session 8))

• tensoring with a line bundle does not change the Brauer equiv. class of twisted lin. system (38 (Session 7))

Then $|\mathcal{O}_P(rd+1)| \sim |\mathcal{O}_P(1)| \sim \mathcal{P}^V$

By Lem 217 (Session 5) and dualizing

(4)

the codim of any twisted lin. sub var of $|\mathcal{O}_P(rd+1)|$
is divisible by $\dim P^{\min} + 1$, therefore $\dim P^{\min} + 1 \mid \deg Z$ #

Corollary 54 P : SB var.

$$\text{per}(P) \mid \text{ind}(P) \mid \text{per}(P)^{\dim P}$$

In particular, the period and index have the same prime factors

pf By Prop 44, $\text{per}(P) \mid \dim P + 1$

Since $\text{per}(P) = \text{per}(P^{\min})$, then $\text{per}(P) \mid \dim P^{\min} + 1 = \text{ind}(P)$

If $\mathcal{O}_P(m)$ is a line bundle, then intersecting the zero set of $\dim P$ -general sections yields a 0-cycle of degree $m^{\dim P}$. For $m = \text{per}(P)$, we have $\text{ind}(P) \mid \text{per}(P)^{\dim P}$ #

Prop 57 P : min SB var.

$$\dim P + 1 = \prod_{\mathbb{Z}} p_i^{c_i} \quad (p_i: \text{distinct primes})$$

then $\exists! P_i$: min SB var s.t. $\dim P_i + 1 = p_i^{c_i}$ and $P \cong \bigotimes_{\mathbb{Z}} P_i$

pf $a := \dim P + 1$, $a_i = p_i^{-c_i} a$.

Write $1 = \sum_i e_i a_i$ and note that $1 \equiv e_i a_i \pmod{p_i^{c_i}}$.

Uniqueness

Assume $P \cong \bigotimes_{\mathbb{Z}} P_i$. By Cor 54, $\text{per}(P_j) \mid \text{ind}(P_j) = p_j^{c_j}$.

hence $p_j^{\otimes e_i a_i} \sim \mathcal{O}^0$ $j \neq i$

Thus $p_i^{\otimes e_i a_i} \sim \bigotimes_{j \neq i} p_j^{\otimes e_i a_i} \sim p_i^{\otimes e_i a_i} \sim P_i$

Then the P_i are unique.

Existence

Set $P_i = (P \otimes e_i a_i)^{\min}$, then

$$P \sim P \otimes \sum e_i a_i \sim \bigotimes_i P_i$$

since P is minimal,

$$P \cong \bigotimes_i P_i \iff \dim P \geq \dim \bigotimes_i P_i$$

Claim $\dim P \geq \dim \bigotimes_i P_i$

By Cor 54 $\text{per}(P) \mid a$, hence $\text{per}(P_i) \mid p_i^{C_i}$

By Cor 54 $\text{ind}(P_i)$ is also a p_i -power

By Lem 51 $\text{ind}(P_i) \mid a$, hence $\text{ind}(P_i) \mid p_i^{C_i}$.

therefore $\dim P_i \leq p_i^{C_i} - 1$, then $\dim \bigotimes_i P_i \leq \dim P$ $\star$